CardioPredictor+

Raspberry Pi installation and remote access guide

This guide takes you from the USB pendrive to a working system on a Raspberry Pi, then to opening the website from another laptop or your phone, including from outside your home. Follow the steps in order. Each command is shown in a box. Type it exactly as shown and press Enter.

Text in boxes is a command for the Raspberry Pi's terminal. Anything marked “replace” must be changed to your own value.

What you need

Item	Details
Raspberry Pi 4 Model B, 4 GB	The server. Use this one. (Raspberry Pi 3 has only 1 GB of memory and is too small for the server.)
Power supply	Official 5.1 V, 3 A USB-C supply for the Pi 4.
microSD card	32 GB or larger, Class 10 or A1/A2 rated.
USB pendrive	Contains CardioPredictor_v4.zip (copied from your laptop).
MAX30102 sensor module	Heart rate and SpO2 sensor, with jumper wires.
Camera	Pi camera module (5 MP or 12 MP) or a USB webcam (see Step 10).
Network	Wi-Fi at home, or Ethernet cable to the router (more reliable).
A laptop with Raspberry Pi Imager	Used once to write Raspberry Pi OS onto the microSD card.

Keep the Pi plugged in whenever the system is running. A sudden power cut can damage the patient database, so use a UPS if you can.

Part A: Prepare the Pi

Step 1: Write Raspberry Pi OS to the microSD card (on your laptop)

1. Download Raspberry Pi Imager from the official Raspberry Pi website (raspberrypi.com, Software page). Install it.

2. Put the microSD card into your laptop (use a card reader if needed).

3. Open Raspberry Pi Imager. Click CHOOSE DEVICE and select Raspberry Pi 4.

4. Click CHOOSE OS. Select Raspberry Pi OS (64-bit) from the list (the version with a desktop is easiest to use).

5. Click CHOOSE STORAGE and select your microSD card.

6. Click NEXT. When asked about customisation, click EDIT SETTINGS and fill in:

- Hostname: cardio-pi

- Username and password: choose a username (for example, clinic) and a strong password. Write them down.
- Wireless LAN: your Wi-Fi network name and password. Set the country to India (or your country).
- Locale: your time zone (Asia/Kolkata) and keyboard layout.
- Services tab: tick Enable SSH and choose Use password authentication.

7. Click SAVE, then YES, then YES again to write the card. Wait until it finishes, then remove the card.

Step 2: Start the Pi and connect to it

1. Put the microSD card into the Pi. Connect the sensor later (Step 8). Connect power.

2. Wait about 2 minutes for the first start.

3. On your laptop, open a terminal (Windows: PowerShell; Mac: Terminal). Connect with:

```
ssh clinic@cardio-pi.local
```

Replace clinic with the username you chose. Type yes if asked about the fingerprint, then enter the password. If cardio-pi.local is not found, find the Pi's IP address in your router's device list and use it instead, for example: ssh clinic@192.168.1.42

4. You are now in the Pi's terminal. All the following steps are typed here, unless the step says otherwise.

Step 3: Update the system

Run these two commands one after the other. The first can take 10 to 20 minutes.

```
sudo apt update && sudo apt full-upgrade -y
```

```
sudo reboot
```

Wait 1 minute, then connect again with the same ssh command from Step 2.

Part B: Copy the project from the pendrive

Step 4: Plug in the pendrive and find it

1. Plug the pendrive into a USB port on the Pi (blue USB 3 ports work too).

2. Look at the name of the drive:

```
ls /media/$USER
```

The output shows the name of the pendrive, for example: SANDISK. Replace PENDRIVE below with that name.

If the list is empty, run: lsblk. Desktop versions usually mount the pendrive automatically. Lite versions may need the drive mounted by hand.

Step 5: Copy and unzip the project into your home folder

1. Install the unzip tool (only needed once):

```
sudo apt install -y unzip
```

2. Copy the zip file from the pendrive (replace PENDRIVE with your drive name):

```
cp "/media/$USER/PENDRIVE/CardioPredictor_v4.zip" ~/
```

3. Unzip it. This creates a folder called CardioPredictor in your home folder:

```
cd ~ && unzip -q CardioPredictor_v4.zip && ls ~/CardioPredictor
```

You should see folders named api, src, models, data, deploy and firmware.

4. Safely remove the pendrive once the copy is finished:

```
sync && sudo umount "/media/$USER/PENDRIVE"
```

Step 6: Check that the Pi has internet access

The installer downloads software, so the Pi must be online. Test it:

```
ping -c 3 pypi.org
```

If you see 0% packet loss, you are online. If not, check the Wi-Fi details in Step 1 or plug in an Ethernet cable.

Part C: Install the system

Step 7: Install CardioPredictor+ (one command, about 20 to 40 minutes)

1. Move into the project folder and run the installer:

```
cd ~/CardioPredictor && bash deploy/install_rpi.sh
```

2. The installer does these things automatically: installs the packages, turns on the I2C port for the sensor, creates the Python environment, trains the model if it is missing, creates the background service, and generates a random dashboard password.

3. When it finishes, it prints a line like this. WRITE THE PASSWORD DOWN:

```
Dashboard password created: aB3xK9...   (write it down)
```

4. If you see an error, copy the last 20 lines of the error and keep them. Common causes are no internet (Step 6) or a power cut during install. Run the same command again: it is safe to repeat.

5. Reboot the Pi. This is needed so the sensor port is switched on:

```
sudo reboot
```

Wait 1 minute, then connect again with ssh.

Step 8: Connect the sensor and check it

Switch off the Pi before connecting or moving wires: run sudo shutdown -h now, unplug, wire, then power on again.

Wire the MAX30102 module to the Pi's pins (use only 3.3 V, never 5 V):

MAX30102 pin	Raspberry Pi pin	Pin number on the Pi
VIN	3.3 V	Pin 1
GND	Ground	Pin 6
SDA	GPIO 2 (SDA)	Pin 3
SCL	GPIO 3 (SCL)	Pin 5
INT	Not connected	Leave empty

Pin 1 is the 3.3 V pin. Find pin 1 by the number printed on the board beside the header. Pins 1 and 2 are the first row, pins 3 and 4 the next row, and so on, all the way to pin 40.

Check that the sensor is seen by the Pi (run this after the reboot):

```
sudo i2cdetect -y 1
```

In the grid you should see the number 57. If you also fitted the optional OLED screen, you will see 3c. If nothing is shown, switch off the Pi, check the wires, and try again.

Step 9: Set the dashboard password and restart

Open the settings file and check the password. The file shows CARDIO_DASHBOARD_PASSWORD=... (write it down):

```
sudo nano /etc/cardiopredictor.env
```

To change the password, edit the line CARDIO_DASHBOARD_PASSWORD=. Save with Ctrl+O, Enter, then exit with Ctrl+X.

Then restart the service so it reads the new password:

```
sudo systemctl restart cardiopredictor
```

Step 10: Choose and test the camera

Choose the camera that matches your hardware:

Your camera	Setting to use	What to do
Camera Module 3 (12 MP) or Module 1 (5 MP), ribbon cable	CAMERA_BACKEND=picamera2	Connect the ribbon to the camera port (CSI). Insert it fully and follow the orientation in the camera's own guide. Use this by default.
USB webcam (any 2 MP, 5 MP or 12 MP USB camera)	CAMERA_BACKEND=usb	Plug into a USB port. Works for any USB camera.
2 MP camera with a small board (OV2640)	CAMERA_BACKEND=usb	Usually cannot be used with the Pi camera port. Use it through a USB adapter board instead.

To change the camera setting, edit the service file (sudo nano /etc/systemd/system/cardiopredictor.service), find the line CAMERA_BACKEND=, change it to picamera2 or usb, save, then run these two commands:

```
sudo systemctl daemon-reload && sudo systemctl restart cardiopredictor
```

Test the camera (it saves a photo into a test patient; you can delete it later):

```
curl -s http://localhost:8000/health
```

The reply should show "sensor":"max30102" when the sensor is connected. If it shows "simulated", the sensor is not found: go back to Step 8.

Step 11: Check that the service is running

The service must show active (running):

```
sudo systemctl status cardiopredictor --no-pager
```

Press q to leave the status screen. If it shows failed, read the last lines of the log:

```
sudo journalctl -u cardiopredictor -n 50 --no-pager
```

Part D: Open the website

Step 12: Open the website on the Pi itself

On the Pi desktop, open the web browser and type this address exactly:

```
http://localhost:8000/dashboard/
```

Log in: enter any username (for example, doctor) and the dashboard password from Step 7 or Step 9.

Step 13: Find the Pi's address on your home network

Run this command and write down the first number shown (it looks like 192.168.1.42):

```
hostname -I
```

This address is used in Parts E and F. Ask your router's settings page to give the Pi the same address every time (called a DHCP reservation, or fixed IP). This stops the address from changing.

Part E: Open the website from other devices on the same Wi-Fi

Step 14: From another laptop or PC

1. Connect that laptop to the same Wi-Fi as the Pi.
2. Open any web browser (Chrome, Edge, Firefox).
3. Type the address below, replacing 192.168.1.42 with the number from Step 13:

```
http://192.168.1.42:8000/dashboard/
```

4. Log in with the dashboard password. The browser remembers the login for this session.

Step 15: From your mobile phone

1. Connect the phone to the same Wi-Fi as the Pi (Settings, Wi-Fi). Do not use mobile data for this step.

2. Open Chrome or Safari and type the same address as in Step 14, for example:

```
http://192.168.1.42:8000/dashboard/
```

3. Log in. For the best view, turn the phone sideways for the charts, or keep it upright and scroll.

4. To open the site like an app, use the browser menu and choose Add to Home screen.

Note: some Android phones can also open http://cardio-pi.local:8000/dashboard/. If that does not work, use the number address.

Part F: Open the website from anywhere (outside your home)

The address in Part E only works on the same Wi-Fi. To open the website from another city, or from mobile data, use Tailscale. It creates a private, encrypted link between your Pi and your devices. Nothing is opened on your home router, and the website is not public on the internet. Tailscale is free for personal use. Create an account once at tailscale.com using your email.

Step 16: Install Tailscale on the Pi

1. In the Pi's terminal, run:

```
curl -fsSL https://tailscale.com/install.sh | sh
```

2. Start Tailscale and sign in:

```
sudo tailscale up
```

3. The Pi prints a web address. Open it on your laptop, sign in to your Tailscale account, and approve the device.

4. Check that the Pi is connected and note its Tailscale address (it starts with 100.):

```
tailscale ip -4
```

5. Also check the Pi's name in the Tailscale admin page (login.tailscale.com, Machines). It is usually cardio-pi.

Step 17: Install Tailscale on your laptop and on your phone

1. Laptop: download Tailscale from tailscale.com/download and install it. Sign in with the same account as the Pi.

2. Phone: install the Tailscale app from the Play Store (Android) or the App Store (iPhone). Sign in with the same account.

3. Turn Tailscale on in both. It shows Connected when it works.

4. Make sure the Tailscale setting called MagicDNS is on (login.tailscale.com, DNS page). This lets you use the name cardio-pi.

Step 18: Open the website from anywhere

On the laptop or phone, with Tailscale turned on, and on any mobile data or any Wi-Fi, open:

```
http://cardio-pi:8000/dashboard/
```

If the name does not work, use the Tailscale address from Step 16 instead, for example:

```
http://100.101.102.103:8000/dashboard/
```

Log in with the dashboard password. The website is now available wherever you have internet access, as long as Tailscale is on and the Pi is powered on.

Step 19: Share access with another doctor or staff member

1. Log in to login.tailscale.com, open the Users page and invite their email address.

2. Once they accept and install Tailscale, they can open the same address from Step 18.

3. Give each person the dashboard password separately, in person or by a secure message. Change the password (Step 9) if it is shared with someone who should no longer have access.

Important: patient share links (Share section on the website) can only be opened by people who are signed in to your Tailscale network. Patients who are not in your network cannot open them. Ask the technical team before enabling a public link for patients, because that needs a different and more secure setup.

Part G: Use the system every day

Step 20: Standard workflow on the website

1. Patient register: choose a patient, or register a new one (ID, name, age, sex). Tick the photo box to take the registration photo.

2. Finger measurement: place a finger on the sensor and press Start measurement. The reading is saved automatically.

3. Risk assessment: the heart rate fills in from the measurement. Enter the other values and press Run risk assessment.

4. Reports: choose PNG, Word (docx) or text, then press Generate and download. Each report includes every reading and the photo.

5. Share: create a link for the patient. It expires automatically after the number of days you choose.

Step 21: Find old data for a patient

Every patient has their own folder, named by the patient ID:

```
ls ~/CardioPredictor/data/patients/
```

Inside each patient folder you will find photos, measurements, assessments and reports.

Example: ~/CardioPredictor/data/patients/P-1042/reports/

Part H: Back up and update

Step 22: Back up the patient data to the pendrive (do this every week)

1. Plug in the pendrive. Check its name:

```
ls /media/$USER
```

2. Copy the data folder (replace PENDRIVE with your drive name):

```
sudo cp -r ~/CardioPredictor/data "/media/$USER/PENDRIVE/cardio-backup-$(date +%F)"
```

3. Safely remove the pendrive:

```
sync && sudo umount "/media/$USER/PENDRIVE"
```

Keep the backup pendrive in a safe place. Patient data is health information.

Step 23: Install a newer version of the software (keeps all patient data)

1. Stop the service and copy the new zip from the pendrive into your home folder:

```
sudo systemctl stop cardiopredictor && cp "/media/$USER/PENDRIVE/CardioPredictor_v5.zip" ~/
```

2. Keep the old data: move the data folder out, unzip the new version, then move the data back:

```
mv ~/CardioPredictor/data ~/cardio-data-keep && cd ~ && unzip -q CardioPredictor_v5.zip && rm -rf ~/CardioPredictor/data && mv ~/cardio-data-keep ~/CardioPredictor/data
```

3. Start the service again:

```
sudo systemctl start cardiopredictor
```

Replace v5 with the version number on your zip file. If the software does not start, read the log in Step 11.

Part I: Troubleshooting

What you see	What to do
Website does not open on other devices	Check the Pi is on, the service is running (Step 11), and both devices are on the same Wi-Fi. For Tailscale, check the app shows Connected on both devices.
Website shows SIMULATED next to Sensor	The sensor is not found. Run sudo i2cdetect -y 1. If 57 is missing, check the wires in Step 8 and reboot.
Sensor says No finger	Place the finger flat over the sensor, press lightly, and keep still for about 10 seconds.

Camera error when taking photo	Check the camera cable or USB camera, then check the CAMERA_BACKEND setting in Step 10.
Report does not download	Read the log: sudo journalctl -u cardiopredictor -n 50 --no-pager
Forgot the dashboard password	Run sudo cat /etc/cardiopredictor.env, then log in with the value after CARDIO_DASHBOARD_PASSWORD=.
Pi does not start the service after a power cut	Run sudo systemctl enable cardiopredictor, then sudo systemctl start cardiopredictor.
The Pi becomes slow	Check memory: free -h. Restart the Pi with sudo reboot. Use a USB SSD instead of the microSD card for better speed.
Install stopped with an error	Make sure the Pi has internet (Step 6). Run bash deploy/install_rpi.sh again. It is safe to repeat.

Part J: Important safety information

- This is a research prototype. It is not a medical device. Results are estimates, not diagnoses.
- The 12-year figure is extended from a 10-year model with a simple assumption. It is an approximation.
- The MAX30102 sensor is hobby-grade. Heart rate and SpO2 are indicative only.
- Blood pressure, cholesterol and glucose are typed in by hand. They are not measured by this device.
- Patient photos and records are health data. Get the patient's consent before storing a photo. Keep the Pi and the pendrive secure.
- Always use a strong dashboard password and do not share it. Turn on Tailscale only on devices you control.

Quick reference

Task	**Command or address**
Connect to the Pi	ssh clinic@cardio-pi.local (replace clinic with your username)
Start, stop, restart the system	sudo systemctl start / stop / restart cardiopredictor
See if it is running	sudo systemctl status cardiopredictor --no-pager
Read the log	sudo journalctl -u cardiopredictor -n 50 --no-pager
Check the sensor	sudo i2cdetect -y 1
Website on the Pi	http://localhost:8000/dashboard/
Website on home Wi-Fi	http://PI-IP-ADDRESS:8000/dashboard/ (from Step 13)
Website from anywhere	http://cardio-pi:8000/dashboard/ (with Tailscale on)
Password file	sudo nano /etc/cardiopredictor.env